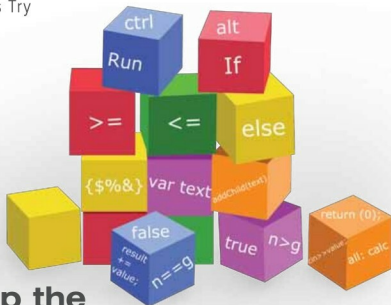




Spicing up the Console for Fun and Profit

Part—1

Command-line terminals and the shell have been an integral part of all *NIX systems from the beginning. The shell is the universal interface between a *NIX system and its users, and the terminal is the medium for it. This article is the first in a two-part series, and shows various ways to spice up different elements of the text console, like prompts, colours or themes to enhance your day-to-day interaction with *NIX systems.

There were different types of hardware terminal devices to interact with the shell in the earlier days of UNIX, but today, these have been mostly replaced by virtual consoles and other emulated terminals provided by various desktop environments. In fact, nowadays, the terminal and shell are just called the text console, mostly due to their mutually complementary relationship. Many different flavours of the shell as well as the terminal were developed in different flavours of *NIX systems. Some popular shells are *bash*, *sh*, *csh*, *ksh*, *zsh* and terminals *xterm*, Linux console, etc. The text console is minimalist, but powerful. Many types of *NIX systems like servers, embedded systems, legacy machines, etc, have neither the need nor the space for resource-hungry, complicated and unreliable graphics environments. Text consoles are the only way to interact with users. Users and developers are still quite dependent on command-line consoles, many decades after the introduction of the first UNIX system.

Over the years, *bash* has emerged as a standard shell on many *NIX systems, especially on GNU/Linux distributions, so the shell used in this article is *bash*. Most modern terminals on GNU/Linux systems are very similar in functionality, so we use the key words 'shell' and 'text console/terminal' interchangeably. Also, I use the word 'terminal' for either *xterm* or the standard GNU/Linux virtual console. I used Puppy Linux 5.1 and the Ubuntu 10.04 64-bit desktop edition

to test code and generate the screen-shots presented here.

The default appearance of the text console is a full-screen virtual terminal in non-graphics mode, or an emulated terminal in graphics mode, with a boring default black background and white text, in most cases. However, the console is highly customisable, based on users' tastes and preferences. Let's find out how.

Tweaking *bash* prompts

Different flavours of GNU/Linux present different kinds of *bash* prompts; the default prompt on my Ubuntu text console and terminal emulator is *user@computername* followed by the current working directory, and \$ for a normal user or # for the super user. You can customise the content and appearance of your prompts to make them more useful and appealing. *Bash* provides the environmental variable *PS1* to customise user prompts. In fact, there are also *PS2*, *PS3*, *PS4* and *PROMPT_COMMAND* environmental variables, relevant to various other aspects of prompt customisation. Just run *OPS1=\$PS1*; *PS1="It's My Prompt :)"* in *bash* to quickly view your prompt change. Restore your old prompt with *PS1=\$OPS1*.

PS2 determines the prompt shown for the continuation of long command lines. It is shown when you break a long command using the \ character, or press *Enter* without completing the shell construct. Try to type *sdemo="any*